TARRIK QUNEIBI

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EXPERIENCE

ISS Corporate (formerly SustainaBase) - *Senior Associate Data Engineer* Feb 2023 - Present

- **Contributed to SustainaBase's successful acquisition by ISS Corporate** by serving as a key technical stakeholder during the due diligence process, participating in executive-level interviews and translating the platform's cloud-native AWS architecture and emissions reporting capabilities into demonstrable business value.
- **Architected and deployed cloud-native AWS Infrastructure-as-Code pipelines** to deliver consistent, audit-ready emissions disclosures and BI tools following environmental reporting standards. Reducing processing cycles from weekly to seconds, accelerating client reporting turnaround.
- **Designed and managed multi-tiered database architecture** integrating relational, NoSQL, and data lakehouse systems to support large-scale data storage, versioning, enrichment, and rapid querying. Enabling both internal analytics and external client reporting across multiple greenhouse gas emission categories.

University of Michigan - *Research Assistant / Project Lead* Jan 2022 - Feb 2023

- **Led a multidisciplinary city-wide water quality research project** coordinating scheduling, field sampling logistics, and task delegation across municipal water treatment facility managers, local businesses, and university research staff. Evaluating UV disinfection's effect on opportunistic pathogens in drinking water distribution systems and presenting technical findings.
- **Conducted comprehensive water quality assessments** including physicochemical characterization, biological sampling, and plant operational data collection across multiple distribution system locations.
- **Established reproducible data management workflows** integrating field measurements, laboratory analysis, and treatment plant operational data into standardized formats with API-driven pipelines, enabling consistent reporting and versioning across the project team.

City of Ann Arbor Drinking Water Treatment Plant - *Water Resource Engineering Intern* May 2021 - Jan 2022

- **Designed and executed an instrumentation validation study** for newly installed post-filtration particle counters. Developing controlled experiments using calibrated particle suspensions to verify detection reliability for Cryptosporidium-sized oocysts and ensure compliance with Surface Water Treatment Rule requirements.
- **Built time-series data processing pipelines** in R to ingest, clean, and analyze continuous particle count streams. Applying statistical methods to correlate filtration performance indicators (turbidity, flow rate, and runtime) with pathogen-surrogate particle breakthrough events.
- **Developed a decision-support dashboard** integrating real-time particle counter data with disinfection contact time parameters, enabling plant operators to make informed, data-driven treatment adjustments in real-time.

SKILLS

Languages: Python, Java, JavaScript, R

Databases & Storage: SQL Server, MySQL, DynamoDB, MongoDB, S3 Data Lake Architecture

Cloud & Infrastructure: AWS Cloud Infrastructure (EC2, S3, RDS, Lambda, Redshift, Stepfunctions)

Domain: Carbon Accounting & Reporting (GHG Protocol, CSRD, CDP), Geospatial analysis (arcGIS), Water modeling (SUMO, Modflow, SWMM), BI Tools (Custom built, Sisense, Tableau)

Machine Learning: Neural Networks, Classification, Gradient Boosted Decision Tree, Linear Regression

EDUCATION

Master of Science in Environmental Engineering - The University of Michigan Dec 2022

Bachelor of Science in Environmental Engineering - The University of Michigan Dec 2021